

AI LOAN ELIGIBILITY CHECKER

AI-Powered BFSI Financial Decision Support Platform

PROJECT DOCUMENTATION

SmartBridge / FutureSkills Prime Student Project

Student Name: _____

College / Department: _____

Academic Year: 2026

1. Abstract

The AI Loan Eligibility Checker is an AI-enabled BFSI (Banking, Financial Services and Insurance) web platform designed to simplify financial decision-making. It combines a Loan Eligibility Checker, Credit Score Analyzer, EMI Calculator and AI Financial Tips in one responsive interface. The frontend uses HTML5, CSS and JavaScript with a dark glassmorphism design. Secure server-side Claude integration and Google Sheets storage are planned as integration stages. The platform is an educational decision-support project and is not an actual lending decision or professional financial advice.

2. Introduction

Financial products can require users to understand income, existing obligations, credit score, interest rates and repayment costs. This project combines common financial utilities into one simple web application so users can explore these concepts through an accessible interface.

3. Problem Statement

Users may find it difficult to estimate loan readiness, interpret a credit score and understand repayment costs when information is spread across different tools. The proposed platform brings these basic utilities together in one application.

4. Objectives

Develop a responsive BFSI web platform; provide a rule-based loan readiness estimate; analyze an entered credit score; calculate EMI, total interest and repayment; provide AI-powered financial guidance; validate user inputs; keep API secrets out of frontend code; deploy the application and document the implementation.

5. Proposed System

The system contains a responsive frontend, financial calculation logic, an AI integration layer and a data-storage integration layer. Users enter information through forms, the application validates it, performs calculations or sends an AI request through a protected backend, and displays the result.

6. Technologies Used

Technology	Purpose	Status
HTML5	Page structure	Implemented
CSS3	Glassmorphism, responsive UI	Implemented
JavaScript	Validation, calculations, interaction	Implemented
Python	Backend support where required	Planned
Git/GitHub	Version control/source hosting	Implemented
Vercel	Web deployment	Implemented
Anthropic Claude API	AI financial guidance	Integration stage
Google Sheets	Structured record storage	Integration stage

7. System Requirements

Hardware: Intel Core i5 10th Gen / AMD Ryzen 5 or equivalent; minimum 8 GB RAM; 256 GB SSD; optional NVIDIA GTX 1650+ for AI/ML experimentation; stable internet connection.

Software: Windows 10/11, macOS or Linux; modern Chrome/Firefox/Edge; VS Code or another code editor; Git/GitHub; appropriate backend tooling.

8. System Workflow

1. User opens the FinPilot AI web application
2. User selects a module
3. User enters required information
4. JavaScript validates the input
5. Financial logic or protected AI backend processes the request
6. Result is displayed
7. Configured records can be stored through the data integration

[USER] → [WEB UI] → [VALIDATION] → [FINANCIAL LOGIC / AI BACKEND] → [RESULT]

9. Project Modules

9.1 Loan Eligibility Checker

Collects age, monthly income, credit score, existing EMI, employment type and requested loan amount. The MVP uses transparent rule-based scoring to estimate financial readiness. Actual lenders may use different criteria.

9.2 Credit Score Analyzer

Accepts a 300–900 score and provides a general educational interpretation such as Excellent, Very Good, Good, Fair, Needs Attention or Low.

9.3 EMI Calculator

Calculates monthly EMI, total interest and total repayment using the standard reducing-balance formula.

9.4 AI Financial Tips

Provides a user-facing interface for financial questions. The production architecture keeps the Claude API key server-side rather than exposing it in browser JavaScript.

10. EMI Formula

$$EMI = [P \times R \times (1 + R)^N] / [(1 + R)^N - 1]$$

P = principal amount, R = monthly interest rate, N = number of monthly installments.

11. User Interface and UX

- Dark glassmorphism design
- Responsive desktop and mobile layout
- Feature cards and clear navigation
- Form validation and readable result cards
- Smooth transitions
- Educational-use disclaimers

12. Security and Privacy

- Never expose the Anthropic API key in public frontend files.
- Store secrets in server-side environment variables.
- Validate inputs before processing.
- Avoid collecting unnecessary sensitive financial information.
- Use HTTPS deployment and protected backend endpoints.
- Clearly communicate that results are estimates.

13. Testing Plan

Test Case	Input/Action	Expected Result	Status
Loan validation	Invalid age/credit score	Validation message	To verify
Loan analysis	Valid financial details	Readiness result	Implemented
Credit analyzer	Score 750	Very Good interpretation	Implemented
Credit validation	Outside 300–900	Validation message	Implemented
EMI	₹500,000; 10.5%; 5 years	EMI and totals	Implemented
AI tips	Valid question	AI response	Integration stage
Responsive UI	Mobile browser	Adaptive layout	To verify
Deployment	Open Vercel URL	Application loads	Implemented

14. Results

The project provides a unified interface for the four planned financial modules. The frontend MVP supports loan readiness analysis, credit-score interpretation and EMI calculations, with an AI tips interface prepared for secure Claude integration. The application is deployed through Vercel for web access.

15. Advantages

- Multiple financial utilities in one interface
- Simple responsive user experience
- Fast client-side calculations
- Transparent rule-based MVP logic
- Extensible AI and storage architecture
- Demonstrates HTML, CSS, JavaScript, Git and AI integration skills

16. Limitations

- Loan eligibility is an educational estimate, not lender approval.
- Credit analysis does not replace an official credit-bureau report.
- AI output can be imperfect and should not be treated as professional financial advice.
- Initial eligibility logic is rule-based rather than a trained ML model.

17. Future Scope

- Authentication and user profiles
- ML-based prediction with a suitable dataset
- PDF financial report generation
- Additional financial and insurance tools
- Improved audit logging and privacy controls
- Analytics dashboard
- Production-grade Google Sheets/database integration

18. Deployment

The source is maintained in GitHub and the web application is deployed using Vercel.

Live URL: <https://fin-pilot-ai-sooty.vercel.app/>

19. Project Repository

GitHub: <https://github.com/somshetwarprasad06-png/FinPilot-AI>

20. Screenshot Placeholders

- Figure 1 – Home Page
- Figure 2 – Loan Eligibility Checker
- Figure 3 – Credit Score Analyzer
- Figure 4 – EMI Calculator
- Figure 5 – AI Financial Tips
- Figure 6 – Mobile Responsive View
- Figure 7 – GitHub Repository
- Figure 8 – Vercel Deployment

21. References

- Anthropic API documentation – <https://docs.anthropic.com/>
- Vercel documentation – <https://vercel.com/docs>
- MDN Web Docs – <https://developer.mozilla.org/>
- FutureSkills Prime / SmartBridge project requirements supplied for the project.

Student Declaration

I confirm that this documentation describes the implementation and planned integrations of the AI Loan Eligibility Checker developed as an educational project.

22. Project Screenshots & Implementation Evidence

The following screenshots document the working FinPilot AI interface, its four core modules, and the connected Google Sheets data-storage workflow.

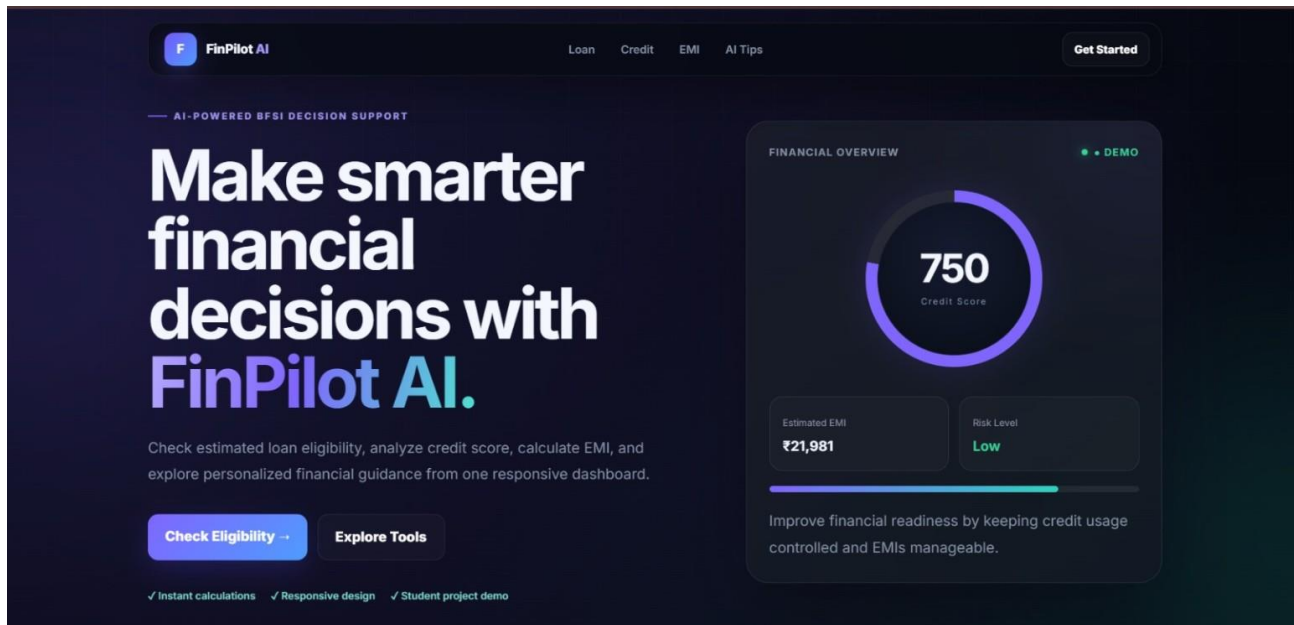


Figure 1 – FinPilot AI Home Page / Dashboard

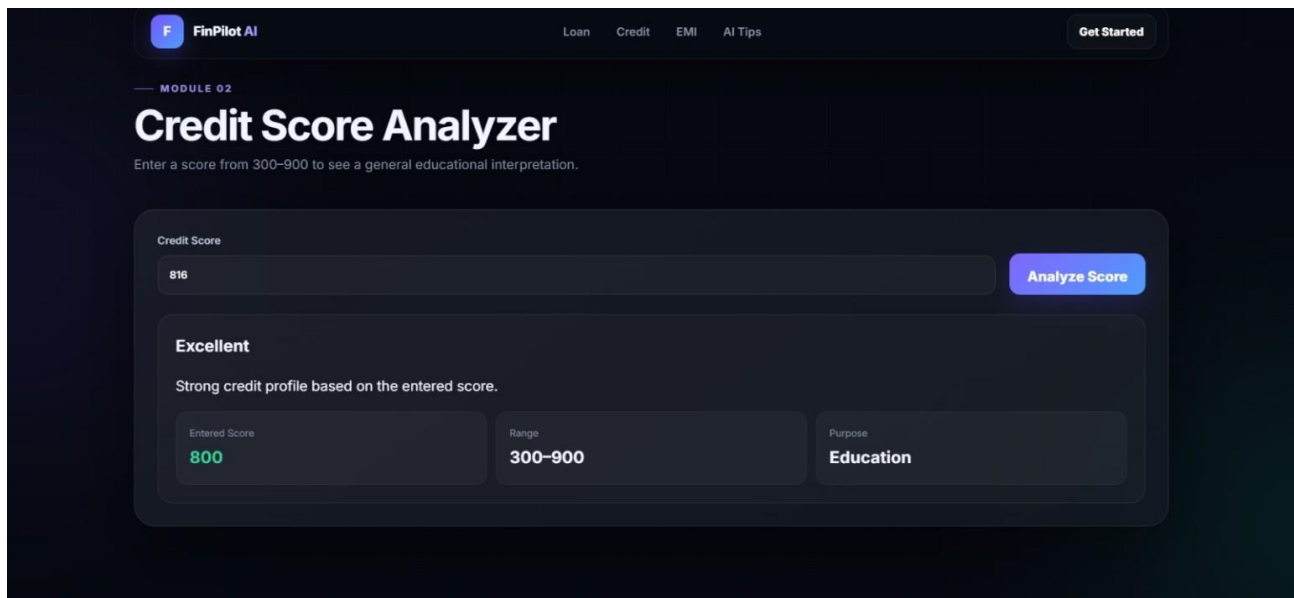


Figure 2 – Credit Score Analyzer

F

FinPilot AI

Loan

Credit

EMI

AI Tips

Get Started

MODULE 01

Loan Eligibility Checker

This is an estimate for educational/project demonstration purposes, not a lending decision.

Age

25

Monthly Income (₹)

250000

Credit Score

800

Existing Monthly EMI (₹)

50000

Employment Type

Other

Requested Loan (₹)

1000000

Analyze Eligibility

✓

Estimated Eligible

This rule-based result is a project estimate and is not an actual lender approval.

Readiness Score

93/100

Risk Level

Lower

Est. Range

₹10,00,000

Debt-to-income ratio: 20.0%. Actual lenders use their own underwriting criteria.

Figure 3 – Loan Eligibility Checker

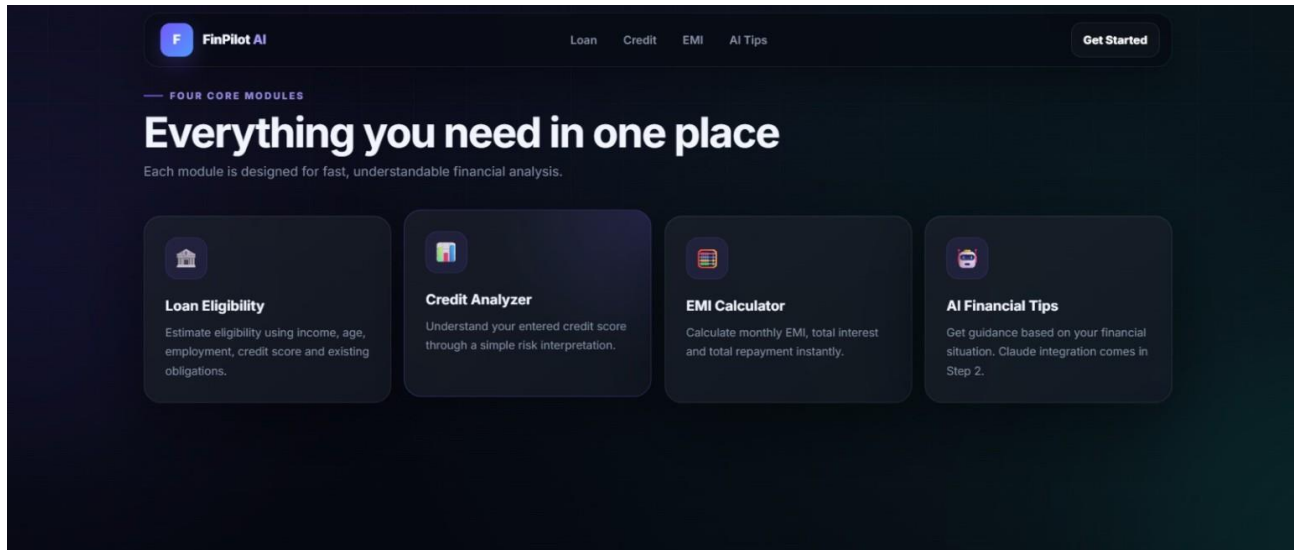


Figure 4 – Four Core Modules Overview

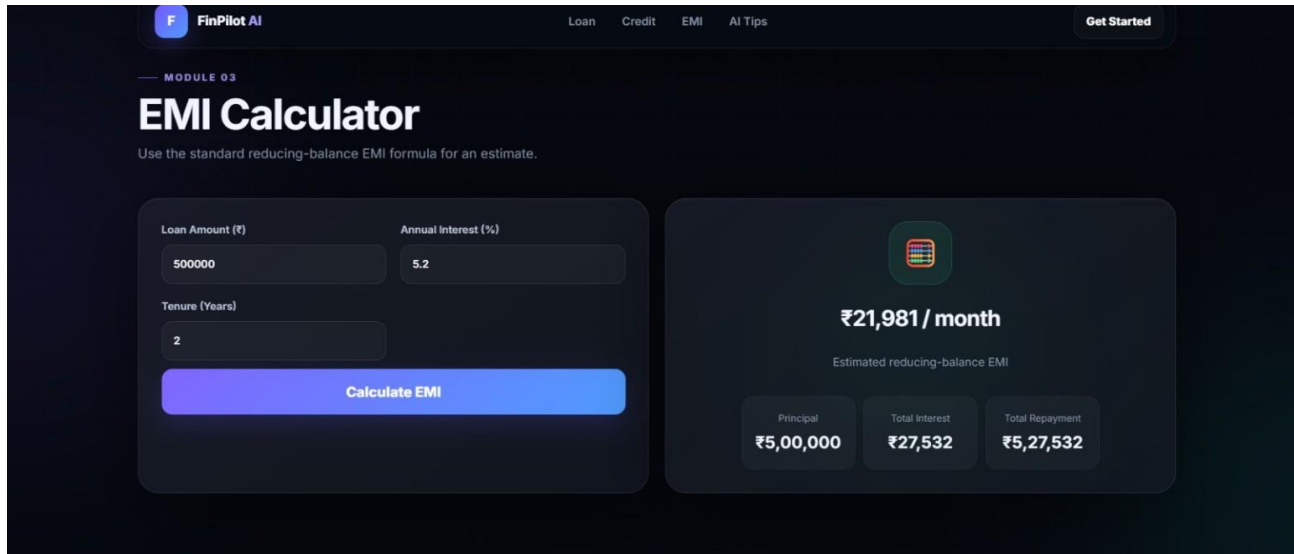


Figure 5 – EMI Calculator

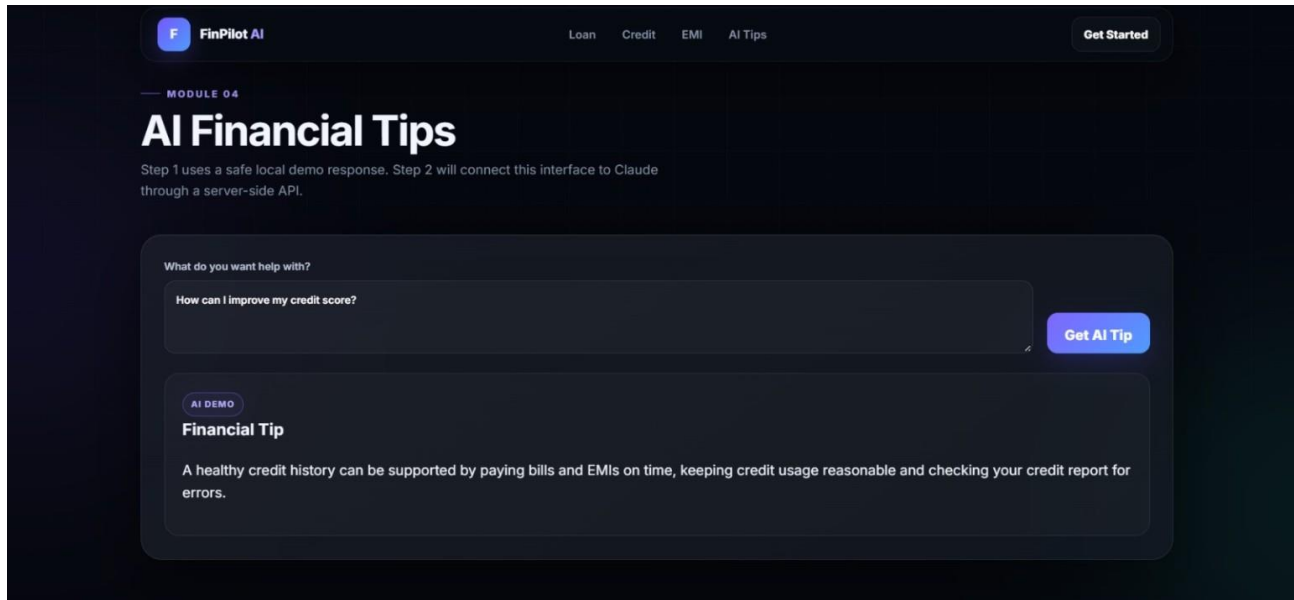


Figure 6 – AI Financial Tips Module

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Timestamp	Age	Monthly income	Credit score	Existing EMI	Loan Amount	Employment	DTI	Result					
2	9/27/2026 15:10	18	50000	750	2500	50000	Student		5	Estimated Eligible				
3	9/27/2026 15:10	18	50000	750	2500	50000	Student		5	Estimated Eligible				
4	9/27/2026 15:16	18	50000	800	2500	50000	Salaried		5	Estimated Eligible				
5	9/27/2026 15:23	18	50000	800	2500	50000	Self-employed		5	Estimated Eligible				
6	9/27/2026 15:32	19	2500	750	1000	5000	Student		40	Needs Improvement				
7	9/27/2026 15:53	25	25000	800	5000	100000	Other		20	Estimated Eligible				
8	9/27/2026 17:36	25	250000	800	50000	1000000	Other		20	Estimated Eligible				
9														
10														
11														
12														
13														
14														
15														
16														

Figure 7 – Google Sheets Financial Record Storage

23. Google Sheets Integration Evidence

The submitted screenshot shows financial records being stored in a Google Sheet with fields including timestamp, age, monthly income, credit score, existing EMI, loan amount, employment, DTI and result. This provides evidence of the project's data-storage workflow.

24. Current Implementation Status

Feature	Status
Responsive FinPilot AI frontend	Completed
Loan Eligibility Checker	Completed
Credit Score Analyzer	Completed
EMI Calculator	Completed
AI Financial Tips interface	Completed / demo response
Google Sheets data storage	Connected / demonstrated
GitHub repository	Completed
Vercel deployment	Completed
Claude API production integration	To be finalized / verify server-side

25. Submission Checklist

- ☐ Project source code available in GitHub
- ☐ Project deployed on Vercel
- ☐ Four required financial modules demonstrated
- ☐ Google Sheets data-storage evidence captured
- ☐ Project documentation prepared
- ☐ Screenshots inserted into documentation
- ☐ FSP registration completed
- ☐ GOI registration completed
- ☐ DigiLocker verification completed
- ☐ Assessment completion verified
- ☐ Project submitted before 30 September 2026

Prepared for academic/project submission – FinPilot AI